Answer Key - Science - K-12 Level Assessment 15 (Answers are ordered exactly as in the question paper.)

# Multiple Choice

1. **Answer:** B
   *Options:* A) building proteins; B) breaking down wastes; C) controlling the activities of the cell; D) converting energy from one form into another
   *Note:* The correct answer is "breaking down wastes" because lysosomes contain digestive enzymes that break down macromolecules and cellular debris, facilitating the digestion of food and the removal of waste products within the cell.

2. **Answer:** A
   *Options:* A) A nail is left outside and it rusts.; B) A glass is dropped and it shatters into small pieces.; C) A rubber band is stretched until it breaks.; D) A pencil is sharpened to a point.
   *Note:* The rusting of a nail is a chemical change that transforms iron into iron oxide, resulting in a different material being formed due to the reaction with oxygen and moisture.

3. **Answer:** C
   *Options:* A) the Moon does not rotate on its axis.; B) the far side of the Moon faces Earth during the day.; C) the Moon makes one rotation per revolution around Earth.; D) the rate of rotation is the same for Earth and the Moon.
   *Note:* The correct answer is based on the concept of synchronous rotation, where the Moon takes the same amount of time to rotate on its axis as it does to orbit Earth, resulting in the same side always facing our planet.

4. **Answer:** D
   *Options:* A) size; B) shape; C) weight; D) hardness
   *Note:* Both the statue and the table are made of the same type of marble, which means they share the same intrinsic material properties, including hardness, regardless of their shape or function.

5. **Answer:** D
   *Options:* A) A bird builds its nest in a cactus.; B) A rattlesnake shakes its tail when threatened.; C) A lizard crawls under a rock for shelter.; D) The ears of a jackrabbit release heat.
   *Note:* The ears of a jackrabbit are large and vascularized, allowing for effective heat dissipation, which helps the animal regulate its body temperature in the extreme heat of a desert environment.

# True / False

6. **Answer:** False
   *Options:* A) True; B) False
   *Note:* Heat energy from the Sun is primarily transferred to Earth through radiation, not conduction, as it travels through the vacuum of space and the atmosphere.

7. **Answer:** False
   *Options:* A) True; B) False
   *Note:* A standardized taxonomic classification system is primarily used to categorize and organize organisms based on their evolutionary relationships and characteristics, rather than to determine their geographical habitats.

8. **Answer:** False
   *Options:* A) True; B) False
   *Note:* The average distance between Earth and the Sun is approximately 149.6 million kilometers, which is correctly expressed as 1.496 x $10^8$ kilometers, not 149.6 x $10^5$ kilometers.

9. **Answer:** True
   *Options:* A) True; B) False
   *Note:* When a battery-operated flashlight is turned on, the chemical energy stored in the battery is converted into electrical energy, which then transforms into light energy through the flashlight's bulb.

10. **Answer:** True
    *Options:* A) True; B) False
    *Note:* Designing an experiment with multiple types of popcorn and repeated trials ensures reliable data collection and minimizes variability, allowing for a more accurate determination of popped kernel yield.

# Long Form Response

11. **Answer:** The formation of a beach surface involves various natural processes, primarily driven by water, wind, and sediment. Waves from the ocean continuously crash onto the shore, eroding rocks and transporting sand and other sediments. Wind plays a significant role as well, moving loose sand along the beach and shaping the dunes through processes like saltation and deflation. Additionally, the type and amount of sediment available, such as pebbles, sand, or silt, influence the texture and appearance of the beach. Together, these elements create a dynamic environment where the beach is constantly changing due to the interplay between erosion and deposition. This ongoing process contributes to the unique characteristics of each beach, making them vital ecosystems.
    *Note:* The answer correctly identifies the interrelated roles of ocean waves in erosion and sediment transport, wind in shaping the beach landscape, and the influence of sediment type on beach texture, capturing the essential processes of beach formation.

12. **Answer:** The study of earthquakes is crucial for understanding Earth's geological history and the processes that shape our planet over time. By analyzing the patterns and causes of earthquakes, scientists can learn about tectonic plate movements, which are responsible for creating mountains, valleys, and ocean basins. Earthquakes also reveal information about the Earth's interior structure and the materials that make up our planet. Additionally, studying historical earthquake data helps us understand how the Earth's surface has changed over millions of years, providing insights into natural disasters' impacts on human civilizations. Overall, earthquakes serve as a key concept in geology, illustrating the dynamic nature of our planet.
    *Note:* correct because it emphasizes that the study of earthquakes provides insights into tectonic plate movements and the Earth's internal structure, which are essential for understanding geological history and the dynamic processes that shape the planet over time.

*End of Answer Key*